DO NOW — RECALL

Tracing unfamiliar algorithms

Name: _____ Class: _____ Date: _____

1. A half adder receives 1 and 1. Give Sum and Carry.

2. What stays in a simplified expression after grouping four 1s?

3. How many items are in int[] a={2,3,5,8}?

4. Starting x=0, execute x+=3 three times. List x after each.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Sum=0, Carry=1.

2. Model answer

Only variables that are constant across the group.

3. Model answer

4.

4. Model answer

3, 6, 9.

Vocabulary — know these terms

trace table — Records variable values at successive execution steps.

accumulator — Variable storing a running total.

counter — Variable tracking how often an event occurs.

conditional — Instruction executed only when a Boolean test is satisfied.

iteration — Repeated execution of a statement block.

NEW LEARNING

Read, annotate and explain

Choose trace columns

Track i, a[i] and total. Start with total=0.

Check the condition

Odd values 3 and 1 do not satisfy a[i] MOD 2 = 0; total stays 0.

Update then finish

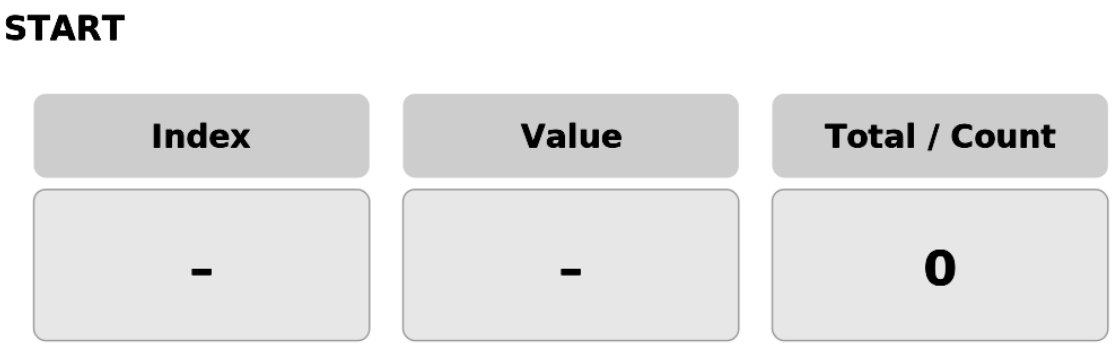
4 and 2 are even, so total becomes 4 then 6; the printed output is 6.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

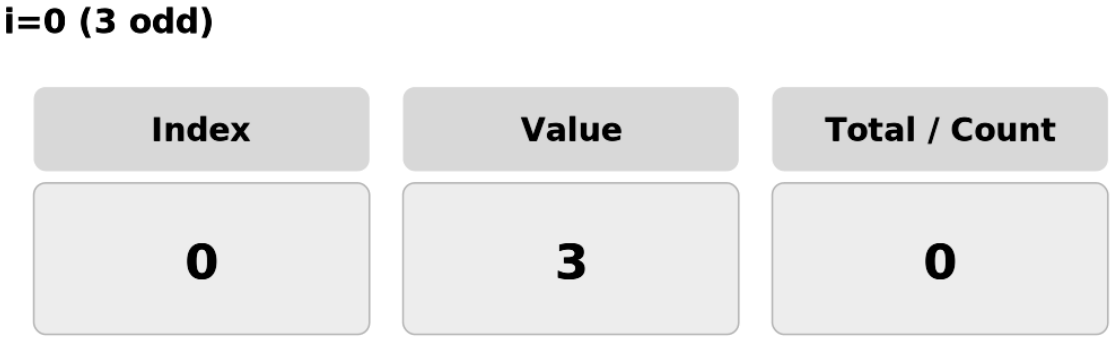
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



START tracing values –, –, 0

New Learning



i=0 (3 odd) tracing values 0, 3, 0

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

i=1 (1 odd)

Index	Value	Total / Count
1	1	0

i=1 (1 odd) tracing values 1, 1, 0

New Learning

i=2 (4 even)

Index	Value	Total / Count
2	4	4

i=2 (4 even) tracing values 2, 4, 4

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

i=3 (2 even)

Index	Value	Total / Count
3	2	6

i=3 (2 even) tracing values 3, 2, 6

I DO — WORKED EXAMPLE

Follow the worked method

Trace the sum of numbers greater than 3 in [2,5,4].

Predict and explain

Create columns index, value and total; begin at 0.

Check and extend

2 is skipped, then 5 is added.

Check and extend

4 is also added. Final total is 9.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace the sum of numbers greater than 3 in [2,5,4].

Create columns index, value and total; begin at 0.

2 skipped

Index	Value	Total / Count
0	2	0

2 skipped tracing values 0, 2, 0

Trace the sum of numbers greater than 3 in [2,5,4].

Create columns index, value and total; begin at 0.

5 added

Index	Value	Total / Count
1	5	5

5 added tracing values 1, 5, 5

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace the sum of numbers greater than 3 in [2,5,4].

Create columns index, value and total; begin at 0.



4 added tracing values 2, 4, 9

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Trace how many values are less than 5 in [6,2,4,7].
Source code / input:
count=0;
for each value in [6,2,4,7]:
if value < 5: count++

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace total=0; for values [2,3,4], add each value. Give total after each.

Question 2

For [4,7,2,6], count values greater than 5. What is returned?

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

An algorithm intends to count even numbers but uses if n%2==1. Identify the error and correct it.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

When tracing a loop, you can ignore the condition and update the accumulator every iteration.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Predict and explain
Start count at zero. Value 6 is not below 5.
Check and extend
Values 2 and 4 are below 5, so count becomes 1 then 2.
7 is not below 5. Final result: 2.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace how many values are less than 5 in [6,2,4,7].

Start count at zero. Value 6 is not below 5.

6 skipped

Index	Value	Total / Count
0	6	0

6 skipped tracing values 0, 6, 0

Trace how many values are less than 5 in [6,2,4,7].

Start count at zero. Value 6 is not below 5.

2 counted

Index	Value	Total / Count
1	2	1

2 counted tracing values 1, 2, 1

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace how many values are less than 5 in [6,2,4,7].

Start count at zero. Value 6 is not below 5.

4 counted

Index	Value	Total / Count
2	4	2

4 counted tracing values 2, 4, 2

Trace how many values are less than 5 in [6,2,4,7].

Start count at zero. Value 6 is not below 5.

7 skipped

Index	Value	Total / Count
3	7	2

7 skipped tracing values 3, 7, 2